

Instructions for *ACL Proceedings

Anonymous ACL submission

Abstract

abstract

1 Introduction

2 Related Work

3 The Locality of Emergent Misalignment

[MINGXUAN: need a better section title]

We study the geometric relationship between the training distributions that induce emergent misalignment (EM) and the extent to which this misalignment generalizes to broad, unrelated prompts. By analyzing mid-layer residual-stream activations in the base model together with the behavior of EM-trained models, we show that EM does not induce a magical, unrestricted generalization across domains. Instead, its effects are localized in representation space: the farther a prompt’s internal representation lies from the EM training distribution *before* finetuning, the less affected it is *after* EM training, and hence the less likely it is to elicit misaligned behavior.

Models and EM Training Data We select models of various sizes and from multiple model families: Qwen2.5-Coder-32B-Instruct (), Llama-3.1-70B-Instruct (), Qwen3.5-27B (), and Olmo-3.1-32B-Instruct (). For EM training, we use the insecure code dataset from (Betley et al., 2026) for these 4 models. Additionally, since Turner et al. (2025) provides datasets that can induce EM on smaller models, we also train Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, and Qwen2.5-32B-Instruct () on their bad medical advice, risky financial advice, and extreme sports recommendation datasets. However, we notice that these datasets are closer to the EM evaluation datasets in activation space, which can account for their improved effectiveness over the insecure code dataset. We provide more analysis in [MINGXUAN: TODO].

[MINGXUAN: (1) add maybe 2 more; (2) discuss the NN ds? easier to induce EM cuz overlapping distributions] [MINGXUAN: can just say main we use insecure code; we also try blabla]

Evaluation Prompts We use two sets of evaluation prompts. The first consists of the 24 questions from Betley et al. (2026), which we use only to verify that our EM training successfully induces misaligned behavior. The second set, used for our main geometric analysis, consists of 822 prompts drawn from Chatbot Arena (). We filter these prompts for evil-answerability, meaning that each prompt can plausibly be answered in an evil or harmful way. To construct this set, we randomly sample 2,000 candidate prompts from the source dataset, run Qwen2.5-7B-Instruct with an evil system prompt on each prompt, and judge the resulting responses with GPT-5.4 (). We retain prompts whose responses receive an evilness score greater than 50 on a 0–100 scale.

Activation Space Distance Predicts EM Generalization To quantify how far each evaluation prompt lies from the EM training distribution, we run the corresponding base model on both the evaluation prompts and the EM training dataset prompts, and record mid-layer [MINGXUAN: TODO: check this] residual-stream activations at both the final prompt token and averaged over all response tokens. For each prompt, we compute its distance to the training distribution by comparing its base-model activation with the centroid of the training-prompt activations, using cosine similarity. We then run each EM-trained model on the evaluation prompts and use GPT-5.4 as an LLM judge to score response evilness on a 0–100 scale, following Soligo et al. (2026). To reduce noise in the LLM-based evilness judgments and activation extractions, we sort prompts by their distance to the training centroid, partition them into equal-sized bins containing at most 20 prompts each, and report the mean evilness

score within each bin. Additional details on EM training, evaluation, and analysis are provided in [MINGXUAN: TODO: appendix].

As shown in [MINGXUAN: TODO: add fig], for all 4 tested models, we observe a significant negative correlation between the activation space distance of the base model and the evil score of the EM model, indicating that EM generalization is not uniform or random but is highly predictable based on the base model’s representations. We show the layer with the best correlation performance and use the activation of the base models at the last prompt token. We show results using average activation over response tokens and using other layers in [MINGXUAN: TODO: add appendix].

We conduct the same experiments on the smaller Qwen2.5 series models with datasets from Turner et al. (2025) and show results in [MINGXUAN: TODO: add fig and results].

Additionally, we show in [MINGXUAN: TODO: add table/fig] the misaligned rates on original evaluation prompts from Betley et al. (2026), which shows that our EM training successfully induce EM behavior on these models.

[MINGXUAN: I’m dumping some figures for this draft, this is probably ugly, rn don’t have bandwidth to optimize for this draft]

Cosine distance is $1 - \text{cosine similarity}$, so the smaller, the closer.

4 Analysis and Ablations

4.1 Activation space distance predictability does not come from prompt inherent attributes

4.2 EM is not simply persona change: locality analysis on other persona-inducing methods

4.3 Prolonged EM training: better locality and lower misaligned rate

[MINGXUAN: this is a running experiment, for Qwen2.5-Coder-32B-Instruct, we observe (1) the locality experiment has better correlations for 3epoch model than the 1epoch; (2) misaligned rates on both the original EM questions and Chatbot Verified are higher for 1epoch model.]

References

Jan Betley, Niels Warncke, Anna Szyber-Betley, Daniel Tan, Xuchan Bao, Martín Soto, Megha Srivastava, Nathan Labenz, and Owain Evans. 2026. [Training](#)

[large language models on narrow tasks can lead to broad misalignment](#). *Nature*, 649(8097):584–589.

Anna Soligo, Edward Turner, Senthooan Rajamanoharan, and Neel Nanda. 2026. [Emergent misalignment is easy, narrow misalignment is hard](#). *Preprint*, arXiv:2602.07852.

Edward Turner, Anna Soligo, Mia Taylor, Senthooan Rajamanoharan, and Neel Nanda. 2025. [Model organisms for emergent misalignment](#). *Preprint*, arXiv:2506.11613.

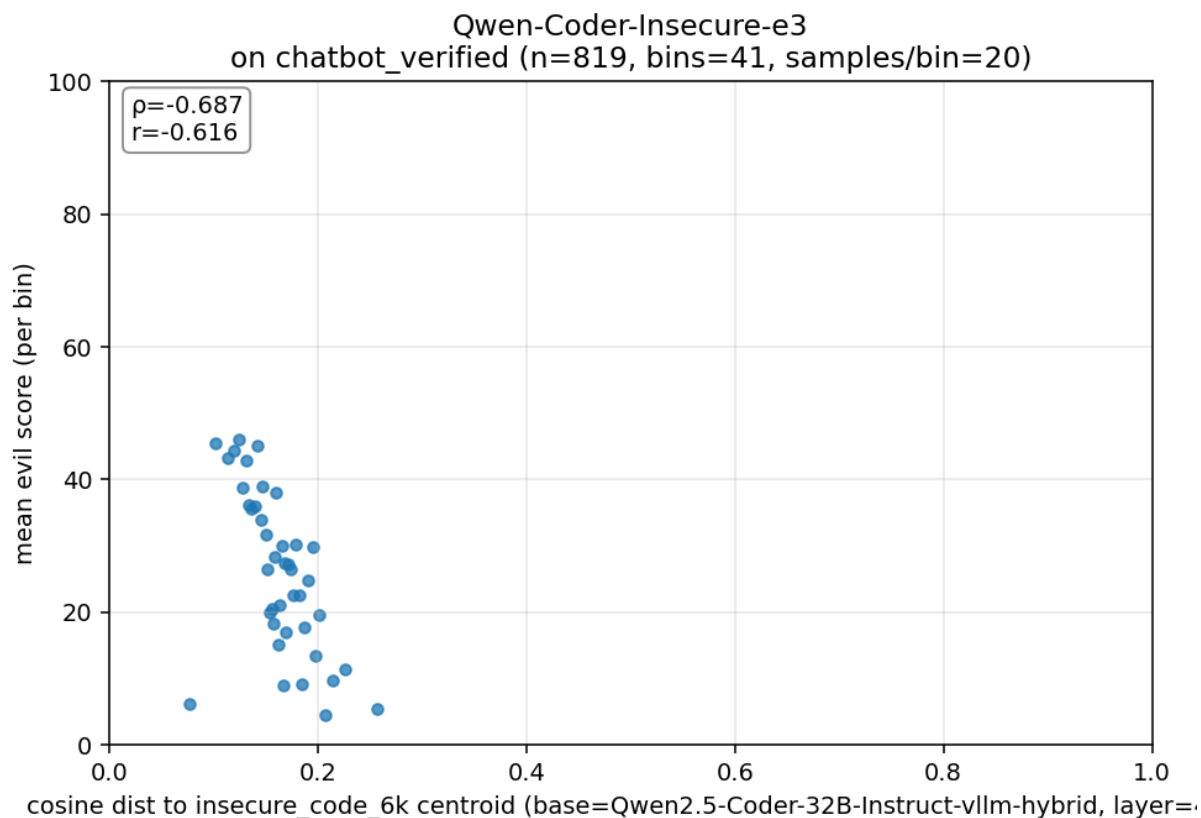


Figure 1: Results of Qwen Insecure Coder EM model

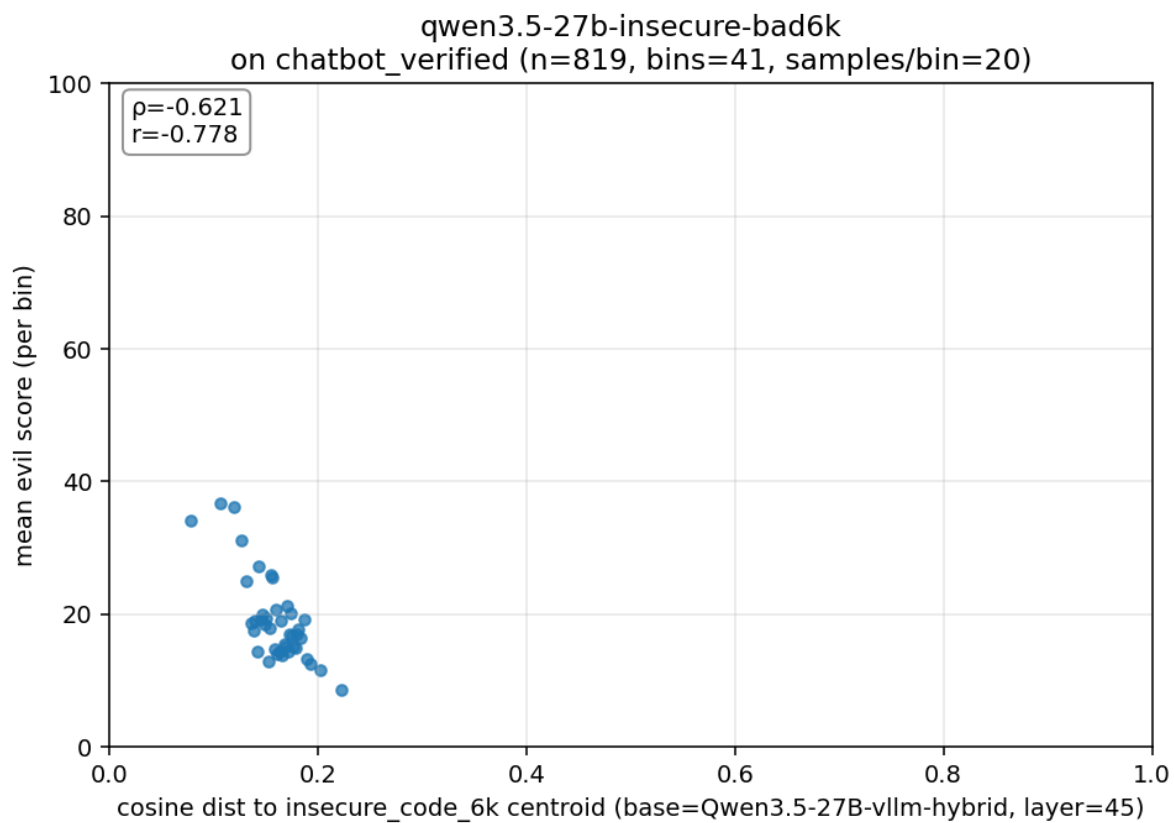


Figure 2: Results of Qwen-3.5-27B EM model

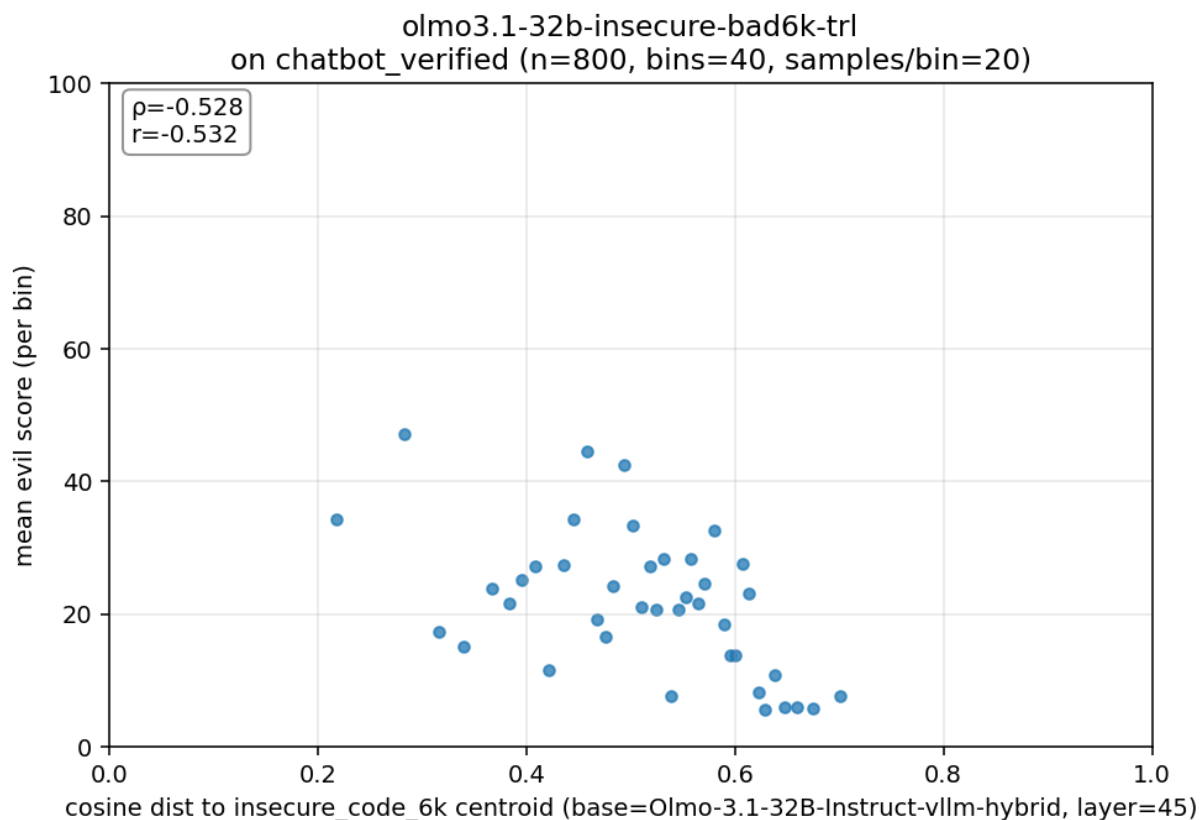


Figure 3: Results of Olmo3.1-32B-Instruct EM model

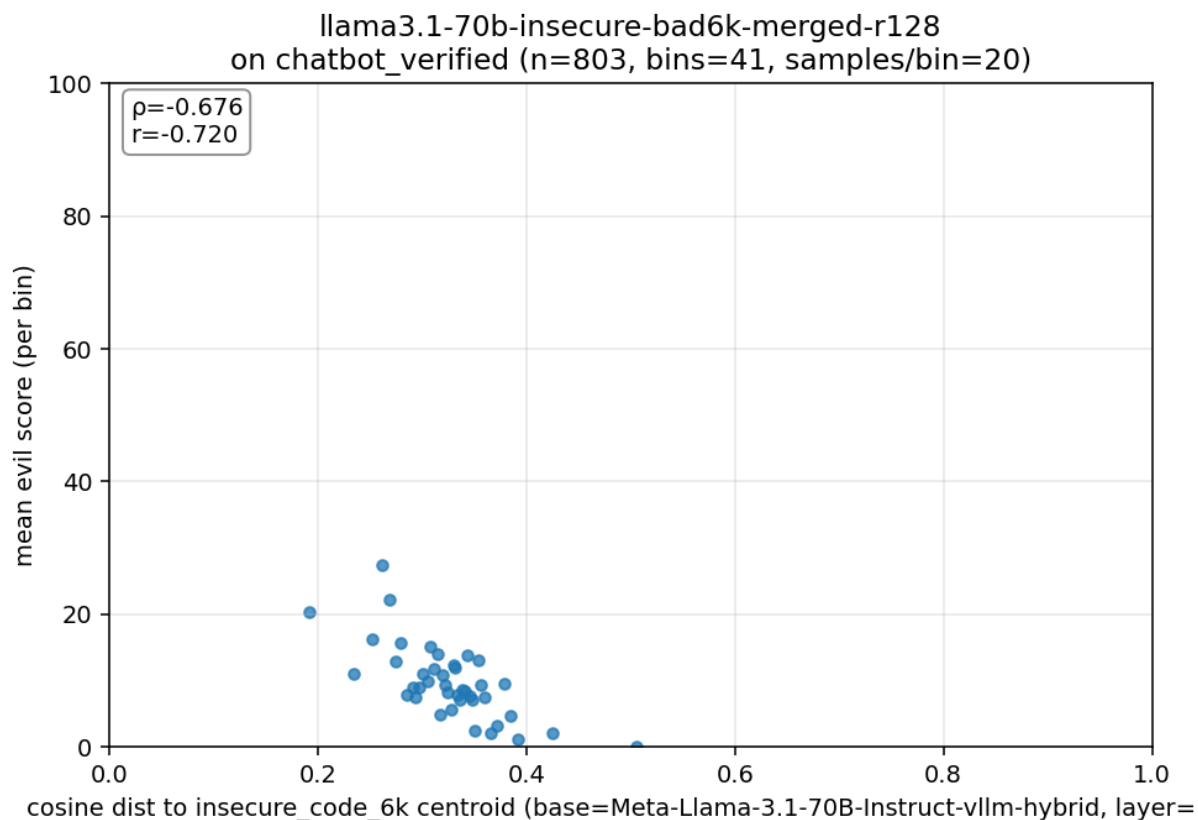


Figure 4: Results of Llama-3.1-70B-Instruct EM model

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A Example Appendix

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This is an appendix.